

Release Notes

Version 1.0.0 - Initial Release

Release Date: 2023-10-15

Status: Stable

Summary

This release introduces the expense management application for our organization, enabling employees to submit, view, and manage their expenses. The application includes RESTful APIs for expense submission, retrieval, approval, and report generation. It also incorporates robust security measures, including JWT-based authentication and role-based access control.

New Features

- **Expense Submission:** Employees can submit their expenses through the API, which are then stored in the database for further processing. This feature enables efficient tracking and management of employee expenses.
- **Expense Retrieval:** Employees and managers can retrieve individual and all expenses through dedicated API endpoints. This feature allows for easy access to expense data for review and reporting purposes.
- **Expense Approval and Rejection:** Managers can approve or reject submitted expenses via API endpoints. This feature streamlines the approval workflow and ensures timely processing of expenses.

Core Capabilities Delivered

- **API Endpoints:** Functional endpoints for expense submission, retrieval, approval, rejection, and report generation.
- **Database Management:** PostgreSQL database for storing employee, expense, approval, and report data.
- **Security:** JWT-based authentication and role-based access control to secure the application.

Improvements & Polish

- **Input Validation:** Custom validation annotations and validators ensure that all input data meets the required criteria.
- **Error Handling:** Standardized error responses and HTTP status codes for consistent and clear error communication.
- **UI Refinement:** Clear and intuitive API documentation for ease of use.

Known Issues & Limitations

- **External Systems Integration:** No integration with external systems in this release. Will be addressed in version 1.1.
- **Performance:** The application handles up to 1000 records efficiently. Performance for larger datasets will be optimized in future releases.

Breaking Changes

- **None:** This is the initial release with no breaking changes.

Compatibility

- **Requires:** Node.js version 14.x, Python version 3.8, PostgreSQL 12.x, Chrome 94+
- **Tested on:** Ubuntu 20.04, Windows 10, macOS 11.x

Upgrade Notes

To deploy this release:

1. Back up the database
2. Pull the latest code: `git pull origin main`
3. Install/update dependencies: `pip install -r requirements.txt && npm install`
4. Run migrations if any: `python migrate.py`
5. Restart services
6. Verify health: `curl http://localhost:8080/api/health`

Contributors

- **John Doe:** Implemented API endpoints and business logic.
- **Jane Smith:** Configured security and input validation.
- **Alex Brown:** Wrote unit and integration tests.

What's Next

Planned for Version 1.1

- **Integration with External Systems:** Integrate with payroll and accounting systems.
- **Performance Optimization:** Optimize database queries for better performance with large datasets.
- **Enhanced Reporting:** Add more detailed and customizable expense reports.

Future Roadmap

- **Mobile Application:** Develop a mobile application for expense submission and approval.
- **Machine Learning:** Implement machine learning for expense prediction and anomaly detection.

Feedback & Support

- **Report issues:** [GitHub Issues](https://github.com/org/expense-management/issues)
- **Request features:** [GitHub Issues](https://github.com/org/expense-management/issues)
- **Documentation:** See [README.md](https://github.com/org/expense-management/blob/main/README.md), [User Guide](https://github.com/org/expense-management/blob/main/docs/user-guide.md), and [API Documentation](https://github.com/org/expense-management/blob/main/docs/api-docs.md)